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Where Minds Pollinate

Class 11
Mathematics

Chapter 4
Exercise 4.9

Notes

National Book Foundation

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Exercise # 4.9

1. A rocket rises 20 ft in the first second, 60 ft in the second second and 100 ft in the third second. If it continues at this rate, how many feet will it rise in the 20th second.

Sol:-

Height in 1st second = $a_1 = 20$ ft

Height in 2nd second = $a_2 = 60$ ft

Height in 3rd second = $a_3 = 100$ ft

So, the progression (arithmetic) will be

20, 60, 100, ...

As, $a_1 = 20$, $d = 60 - 20 = 40$

also, $a_n = a_1 + (n-1)d$

for $n = 20$

$$\therefore a_{20} = 20 + (20-1)(40)$$

$$a_{20} = 780 \text{ ft}$$

Therefore, rocket will rise 780 ft in the 20th second.

2. On the results declaration day, the school wants to invite parents as well as students. Auditorium has 21 seats in the first row and each of the other rows has one more seat than the one in front of it. There are 30 rows of seats in total. If they anticipate that 1,200 people will come that day, will there be seat for everyone? Justify your answer.

Sol:-

Seats in 1st row = $a_1 = 21$

Seats in 2nd row = $a_1 + 1 = 22$

Seats in 3rd row = $(a_1 + 1) + 1 = 23$

So, the progression (arithmetic) will be

21, 22, 23, ...

Here, $a_1 = 21$, $d = 1$,

As, $a_n = a_1 + (n-1)d$

for $n = 30$ rows

$$\Rightarrow a_{30} = 21 + (30-1)(1)$$

$$a_{30} = 50 = a_n$$

$$\text{As, } S_n = \frac{n}{2} [a_1 + a_n] \Rightarrow S_{30} = \frac{30}{2} [21 + 50]$$

$$S_{30} = 1065$$

No, the given or provided seat are 1065 but people are 1200, so, these seats are not enough for people coming.

3. Majid retired after 30 years of employment. If his salary was Rs. 4,500 in first year and he received an increment of Rs. 820 at the end of each year of service. What was his total salary after 30 years.

Sol:-

Salary in first year = $a_1 = 4,500$

Common difference = $d = 820$

Total service years = $n = 30$

As, $S_n = \frac{n}{2} [2a_1 + (n-1)d]$ → for arithmetic series

$$S_{30} = \frac{30}{2} [2(4500) + (30-1)(820)]$$

$$S_{30} = 4,91,700$$

So, the employ earned Rs. 4,91,700 in 30 years of his service.

4. You save Rs. 1 in the first day. Then each day thereafter, save double the amount you saved the day before. Find the amount you should save in the 20th day of your plan.

Sol:-

Money saved at 1st day = $a_1 = 1$

Common ratio = $r = 2$

Money save at 20th day = $a_{20} = ?$

It will be a geometric series

$$\therefore a_n = a_1 r^{n-1}$$

for $n = 20$

$$\Rightarrow a_{20} = 1(2)^{20-1}$$
$$a_{20} = 2^{19}$$
$$a_{20} = 524,288$$

Therefore, Rs. 524,288 will be saved at 20th day of the plan.

5. A vacuum pump removes $\frac{1}{5}$ of the air from a sealed container on each stroke of its piston. What percent of the air remains after five stroke of the piston.

Sol:-

$$\text{Air left after 1st stroke} = a_1 = 1 - \frac{1}{5} = \frac{4}{5}$$

$$\text{Common ratio} = r = \frac{4}{5}$$

$$\text{As, } a_n = a_1 r^{n-1}$$

$$a_5 = \left(\frac{4}{5}\right) \left(\frac{4}{5}\right)^{5-1}$$

$$a_5 = 0.3276 \text{ or } 32.76\%$$

6. Aslam borrows Rs. 20,000 at 11% interest compounded annually. If he pay off the loan in full at the end of four years, how much does he pay.

Sol:-

$$\text{Total amount} = 20,000$$

$$\text{Interest applied annually} = 1 + \frac{11}{100} = \frac{111}{100}$$

So, amount done after one year

$$\text{i.e. } a_1 = 20,000 \left(\frac{111}{100} \right)$$

$$\text{Therefore, } a_n = a_1 r^{n-1}$$

$$a_4 = 20000 \left(\frac{111}{100} \right) \left(\frac{111}{100} \right)^{4-1}$$

$$a_4 = 30,361.408$$

7. A property dealer estimates that a piece of a land will increase its value at a rate of 10% each year. If the original value of the land is Rs. 450,000. What will be value in 8 years?

Sol:-

$$\text{Value of Land} = 450,000$$

$$\text{Increase in value each year} = r = 1 + \frac{10}{100} = 1.1$$

$$\text{Value of land at the end of 1st Year} = a_1 = 450,000 (1.1)$$

$$\text{As, } a_n = a_1 r^{n-1}$$

$$a_8 = (450,000)(1.1)(1.1)^{8-1}$$

$$a_8 = 964,614.965$$

8. A man deposits in a bank Rs. 2,000 in the first year, Rs. 4,000 in second year, Rs. 8,000 in the third year and so on. Find the amount he will have deposited in the bank by the fifth year?

Sol:-

Amount deposited in 1st Year = $a_1 = 2,000$

// // // 2nd Year = $a_2 = 4,000$

// // // 3rd Year = $a_3 = 8,000$

The given data show geometric progression with $r = 2$

$$\text{As, } a_n = a_1 r^{n-1}$$

$$a_5 = (2,000)(2)^{5-1}$$

$$a_5 = 32,000$$

9. The no. of bacteria in a culture increased geometrically from 16000 to 121500 in 5 days. Find the rate of increase assuming the rate to be constant.

Sol:-

Initial population = $a_1 = 16,000$

after first day = a_2

after 2nd day = a_3

after 5th day = $a_5 = 121,500$

$$\text{As, } a_n = a_1 r^{n-1}$$

$$a_5 = (16000) r^{5-1}$$

$$\frac{121500}{16000} = r^4$$

$$r = \frac{3}{2}$$

10. A car loan is in the amount of RS. 600,000 from the bank. Interest is 9% compounded annually and the entire amount is to be paid in 10 years. How is to be paid back?

Sol:-

$$\text{Amount} = 600,000$$

$$\text{Interest annually} = 1 + \frac{9}{100} = 1.09 = r$$

$$\text{Amount paid at the end of } 1^{\text{st}} \text{ year} = a_1 = (600,000)(1.09)$$

$$\text{As, } a_n = a_1 r^{n-1}$$

$$a_{10} = (600,000)(1.09)(1.09)^{10-1}$$

$$a_{10} = 1,42,0418.205$$

11. Zain bought a new car and got policy from insurance company. He will pay 5,000 the first year, 6,125 the second year, 7,250 the third year and so on for 10 years. How much he will pay to insurance company.

Sol:-

Amount paid in 1st year = $a_1 = 5,000$

Amount paid in 2nd year = $a_2 = 6,125$

Amount paid in 3rd year = $a_3 = 7,250$

The data shows arithmetic progression with $d = 1125$

$$\text{As, } S_n = \frac{n}{2} [2a_1 + (n-1)d]$$

$$S_{10} = \frac{10}{2} [2(5,000) + (10-1)(1125)]$$

$$S_{10} = 100,625$$

12. Naveed takes a vehicle from bank after paying down payment. He deposits Rs. 13000 in a bank in first month, Rs. 14500 in second month, Rs. 16000 in the third month and so on. Find total amount of deposit he has to pay at the end of two years.

Sol:-

Amount deposited in 1st year = $a_1 = 13000$

Amount deposited in 2nd year = $a_2 = 14500$

Amount deposited in 3rd year = $a_3 = 16000$

The data shows arithmetic progression with $d = 1500$

$$\text{Ans. } S_n = \frac{n}{2} [2a_1 + (n-1)d] \quad \{ 2 \text{ years} = 24 \text{ months} \}$$

$$S_{24} = \frac{24}{2} [2(13000) + (24-1)(1500)]$$

$$S_{24} = 726,000$$

13. A man borrows a loan Rs. 1,000,000 for leasing a car and agrees to repay with a total 20 installments. Each installment is less than the preceding by Rs. 2,000. What is his first installment?

Soln

$$\text{Total money} = 1,000,000$$

$$\text{No. of installments} = n = 20$$

$$\text{Common difference} = d = 2,000$$

$$\text{Ans. } S_n = \frac{n}{2} [2a_1 + (n-1)d]$$

$$S_{20} = \frac{20}{2} [2a_1 + (20-1)(2000)]$$

$$1,000,000 = 10 [2a_1 + 38000]$$

$$-38000 + 100,000 = 2a_1$$

$$a_1 = 31,000$$

14. Sara pays her first installement Rs. 8000 to insurance company for the vehicle. Each installement will increase by 5%. What total amount will she pay in 24 installements?

Sol:-

first installement $= a_1 = 8,000$

second instalment $= a_2 = 8,000(1.05)$

Third instalment $= a_3 = 8,000(1.05)(1.05)$

As,
$$S_n = \frac{a_1(r^n - 1)}{r - 1}$$

$$S_{24} = \frac{8,000[(1.05)^{24} - 1]}{1.05 - 1}$$

$$S_{24} = 356,015.991$$